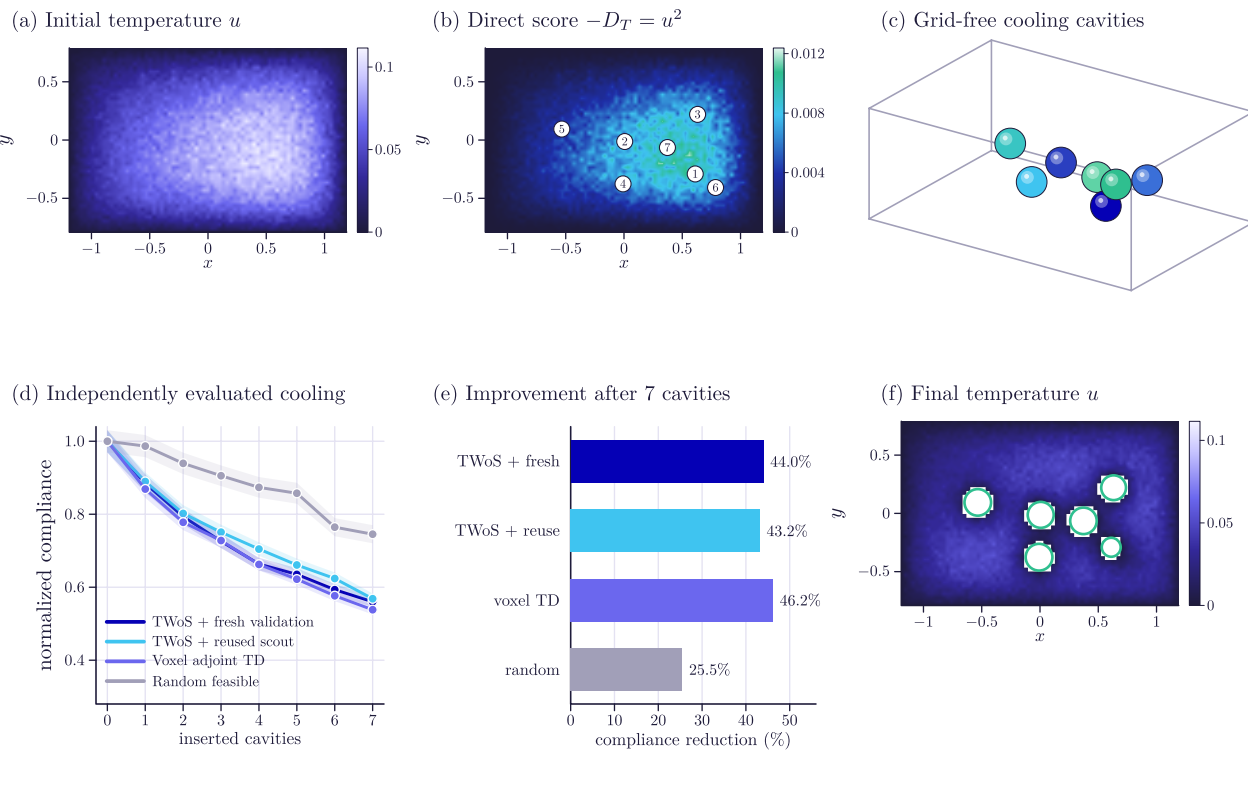

(a) Initial temperature u
(b) Direct score $-D_T = u^2$
(c) Grid-free cooling cavities
(d) Independently evaluated cooling
(e) Improvement after 7 cavities
(f) Final temperature u
x
y
normalized compliance
inserted cavities
TWoS + fresh validation
TWoS + reused scout
Voxel adjoint TD
Random feasible
TWoS + fresh
TWoS + reuse
voxel TD
random
44.0%
43.2%
46.2%
25.5%
compliance reduction (%)